

ActInf Textbook Group ~ Cohort 7 ~ Session 7 (Chapter 4) ~ 9/16/2024

TextbookGroup/ParrPezzuloFriston2022/Cohort_7 | Cohort 7 ~ Session 7 (Chapter 4) ~ 9/16/2024 | 57:02

okay it's September 16th 24 and Andrew is going to give an overview of chapter four then we'll see what questions we have slash other questions already written so go for it

Andre thanks Daniel uh yeah so again this is chapter four active inference uh 2022 textbook uh we're working with cohort 7 so this is the first week of going through chapter 4 so I figured I'd give uh an overview of the chapter uh we're already told off the bat in uh section 4.1 the introduction uh that we're going to be introduced to the relationship here between free energy and bayesian inference uh the form of typical active inference generative models that we often see in active inference implementations uh and we'll also look at the Dynamics obtained from minimizing these models free energy so specifically we look here at both discrete and continuous time models as well as the neurobiologically motivated idea of inferential message passing underlying the computations and predictive coding involved in active inference um so again this is this is the first half of the textbook so this is much more more Theory rather than direct implementation however chapter 4 is particularly uh mathematically involved uh in comparison to the prior three chapters um of course bases rule figures

very strongly here so we see that in section 4.2 from basian inference to free energy um so we recall that we're working with approximate inference rather than exact basian optimality uh given the problem of tractability or intractability uh as well as neurobiological grounding of uh of limited resources so the idea being that for biologically plausible models we have to actually consider uh any kind of um computational limitation of resources or metabolic and other biological uh limitations on the processes that that proceed during pre- energy minimization so that's the kind of matching between biological plausibility and why we're looking at approximate rather than exact uh uh free energy minimization so the uh or rather surprise minimization the mathematics involved here are drawn from linear algebra uh differentiation and calculus and the tailor series expansion uh the textbooks appendices give further details on the relationships between these fields so I strongly REM uh recommend looking at those uh if you if you are still kind of learning the maths involved uh we're shown in equation 4.1 bases rule uh so here this this helps us to better understand a generative model and active inference uh which is a generative model can be described as uh here we can see it as like the probability of x times the probability of Y conditioned on X and then we can flip those around on the other side of the equal sign uh and so that that gives

us a way of understanding like we can we
can use B's rule to potentially update
any aspect or any component of B's role
uh whenever something new is updated
just kind of uh sort of plug and chug
for the rest so to speak right so that
means whenever we have like new
observations uh come in so our model our
our our active inference agent observes
a new observation we can update its uh
posterior beliefs about States so
um so

part of how we're able to do this also
this is kind of a key aspect to
understanding what free energy is
relative to surprise and a lot of the
other theoretical terms that were
introduced to um Jensen's
inequality renders optimization here
possible since minimizing surprise
itself would be next to Impossible it's
intractable whenever it comes to like
very complex models uh integrals that
that Things become rather difficult to
keep track of or there are limited
computational resources whenever doing
the kind of computations involved with
numerous or highly complex matrices um
we we can use Jensen's inequality which
is uh this idea that the log of an
average is always greater than or equal
to the average of a log um it's you know
I don't want to call it a trick but in a
in a sense it's what allows us to say oh
this uh this
tractable uh variational quantity we
call free energy is always greater than
or equal to surprise um that we can see

how that math is played out in equation 4.2 on page 65 so the idea is that uh free energy becomes this tractable quantity that we can calculate uh it it becomes an upper bound on surprise so well if it's the case that we can't perfectly compute surpris in all cases we can compute free energy and so if we know that we can reduce that free energy as much as possible uh it's it's going to close in that that bound that that bound is going to move further and further down closer to to zero such that you know if if we can ideally we'd want free energy to be zero and hopefully that's a good approximation of surprise being also being zero it necessarily would have to be so um

in equation 4.3 we're also introduced to kind of a broader um way as well as equation 4.4 uh how to actually compute pre energy and so we can see that that's a combination of two terms a KL Divergence or koback leeler Divergence which is a way of kind of quantifying the difference between two distributions how different are they from one another uh and so that's that's why the Divergence term has these two different uh terms within it right it's the difference between our variational posterior Over States Q of x uh the the difference between that and the um uh probability of X condition on y so so it's a way of kind of finding the difference we can minimize that as much as possible then all we're left with is

negative log model evidence which itself is kind of a proxy for surprise so if we can minimize the Divergence as much as possible then we come very close to also uh minimizing surprise so um section

4.3 I'm just going to make a couple jumps here because it's the rather broad chapter um section four 4.3 were introduced to uh Factor graph notation for static perception models first uh which are kind of like a precursor to eventually get getting into active inference uh in static perception models we're we're just dealing with perception call that active inference is a combination of action uh uh perception and action you change your mind or you change the world and so without action we're just left with this kind of static perception right so it's this kind of model that updates based on new data but it doesn't itself act it doesn't do anything to change the world and and therefore it also lacks many other components uh in it in its architecture that you would find in a full active inference model um so it gives us examples as to how a prior and likelihood the core components of a generative model are involved in the the model computation uh we're kind of explained like in terms of like binary classification how we can look at a static perception model through positive and true negative rates folks who are

familiar with the kind of machine

learning not even necessarily machine

learning

uh but but that that kind of logic is

very much at play there um and then in

figure

4.3 we're then introduced to active

inference models specifically in figure

4.3 there are uh discrete and continuous

models uh where a temporal Dimension has

also been added where states are

conditionally dependent on the state at

the prior step that is the model carries

its own beliefs about how States

transition over time so um here actions

and policies policies are

considered to be sequences of individual

actions are introduced and as I

mentioned State transitions are

introduced the agent actions derived

from their policies impact the

environment and observations they

receive over time so as we recall agents

are self-evidencing so what they

will do is that they have a belief about

how to go from state to state via acting

and then by acting they're going to

choose actions that hopefully will

elicit the observations they want to see

and we've all uh already been introduced

to the idea of preferences just like a

prior over observations so the idea is

you you you expect to see or not

necessarily see but observe particular

observations and so you act in order

to eventually make those happen so if

someone uh you know is hungry and they

want food or they expect food then they

act in order to obtain it you know and eat it and so on so it's uh it's this one of the most interesting kind of uh aspects to the framework of active inference is the is the self-evidence thing but it it in fact plays out in the model logic um as we look at it here in this chapter so to get further into those components uh section 4.4 goes more into specifically discrete time models where we're looking at specifically partially observable markov decisionmaking processes where as we know from previous chapters the agent has no direct no direct access to the true states of the world and can only infer them um so the agent will attempt to infer them through free energy minimization over various model components so we in a PDP in this case uh we can map the components we've covered previously to components of the pp uh A is our likelihood model which we've already seen in the equations describing a generative model um B is our transitions model I already touched on that it's the it's a it's a distribution over what is the next state going to be given what the current state is as well as an action that the agent can take now basically what is going on now and what should I do in order to get to a more preferable State um and then D is our Vector of Prior overstates so what does the agent believe at the onset uh that's a it's sort of like the initial belief that's kind of right for being updated and then uh as I already

mentioned earlier we have preferences or the C Vector over expected observations um we're also introduced to G as a Quant to be minimized uh it's not exactly equivalent to variational free energy which is often denoted F because G is kind of like the uh free energy of the future or expected free energy and that's where it's it's within that competition that c actually gets directly included so it's kind of like it it allows agents capacity for planning that is they actually incorporate what they want to see into uh kind of their policies and ways of inferring how they should act uh it's it's really in G that that happens so we can think of G as as like kind of planning towards the future meanwhile F variational pre energy is minimized sort of Moment by moment you know um we can also further decompose G so as we saw earlier F to be uh decomposed into like a Divergence term in a negative log model evidence term so with G this can be broken down into actually many ways uh one of them is info gain or Information Gain or epistemic value all those are equivalent uh and pragmatic value and these are probably this is one of the most uh interesting to me personally one of the most interesting aspects to active inference which is that uh G takes account of not just how can an agent realize its goals in terms of pragmatic value which can also kind of be likened to the word utility as it's used in economics or or Game Theory

um but also Information Gain which is an agent actually will act to reduce uncertainty so part of that involves engaging in exploratory behaviors where the agent isn't necessarily trying to immediately uh realize a goal it has instead it might be seeking more information about say the environment that it's in you know so it's kind of like if you if you plan to to fulfill a project like uh you know you could sit down and try and do the project but if you're not sure what kind of research you need to do yet uh and you skip that those preliminary stages of seeking more info uh in order to realize your goal of finishing the project then you might end up creating a very subpar project right and so so all of that plays into kind of classic uh in reinforcement learning and other domains like how do you balance or even Define uh exploratory Behavior with exploitative behavior um exploitative meaning directly realizing goals which is also relatable to that pragmatic value term that we saw expected free energy so the idea of expected free energy part as an answer to that kind of explore exploit dilemma is that you the idea is you can have both you can have both of those included and Via uh expected free energy minimization the agent kind of naturally balances those two things it doesn't mean that they're always going to be literally equivalent it doesn't mean like 0.5 for one and 0.5

for the other it just means that uh the way that this is commonly done in say reinforcement learning is that the programmer has to uh choose some way of defining what exploratory behavior is and more often than not it'll be through some kind of uh arbitrarily chosen uh rule such as Epsilon greedy Behavior which is where you say okay the agent will try to fulfill its goal with a probability of 0.9 it'll pick an action uh that it believes best to fulfill its goal and realize that pragmatic value of 0.9 uh but with the probability of 0.1 it'll just do something random right um and then that becomes equated to exploratory Behavior active inference is is is very different right because you don't have to do that uh at all the the modeler doesn't have to make a definition of what exploratory behavior is it's the agent who sort of autonomously decides what is exploratory behavior and it's not necessarily doing it randomly either right that's another true aspect so um forgive me if that's a lot but I'm personally interested in that that side of active inference so and I think it's very valuable uh in in applying active inference to other fields that more typically do try to resolve that that explore exploit explore exploit uh dilemma but don't always necessarily have an answer for how to go about it um to wrap up uh the some other topics that were introduced to or at least uh that are discussed or brought up in this chapter are uh marov

blankets separating the agent from its environment or a system from uh as Mark Stone says the not system uh right so it's it's a it's a way of finding statistical independencies such that you can actually Define a thing and kind of separate it out from the environment around it which is what allows us to kind of delineate like here's where free energy is happening it's happening within this agent and not U you know it's internal to this this particular system right

um we're also some other topics include generalized coordinates of motion um that'll come up much more in chapter eight future so I won't go into it but uh that's used more expressly for continuous time models and that's sort of how the free energy uh computation plays out so a lot of it just relies upon ideas like Taylor series expansions right so as you as you add more terms you can potentially become increasingly more accurate at at kind of predicting what direction uh certain quantities will move going forward and that's what allows for the agent to start to uh making these kind of continuous time predictions and then update its beliefs accordingly um basy and message passing uh mess passing which uh which we'll figure much more in chapter 5 um but we we just kind of get a hint of it right now this idea yeah um passing messages along cortical layers which can be represented in a a broad variety of ways um you can you can represent it more uh

distinctly as like these are cortical
micro circuits or you can represent them
as more simply like nodes and edges uh
on a factor graph and imagine it like a
a message passing kind of network um the
the active inference Institute we we
working with Folks at bias labs for
example who have a very interesting way
of of representing uh message passing on
on Factor graphs or what they call
constraint forny Factor graphs um it's a
really nice kind of uh ning clature and
and and generally quick way of like
um sort of normalizing how we represent
these kinds of things because it allows
for further work but in any case U
message passing yeah has messages being
passed different kinds of local
computations being formed passed around
so kind of reminds of uh communication
in the brain you know our uh our neurons
are not all uh fully connected uh we'd
presumably be very overloaded if that
were the case so we're dealing with
these kind of you know sparse networks
that uh you know pass messages to each
other in kind of certain kinds of
distinct ways um that you know at a
broader scale allows us to update our
beliefs following the active inference
framework so um I'll leave things there
uh sorry I realized now that that intro
was much longer than it than I planned
but uh I'll open the floor if anyone
wants to to discuss anything or has any
question questions or we can go into any
particular
topics thank

you yeah as you pointed out it it's
a interesting chapter with a mix of both
seemingly General mathematical points to
make like base theorem Jensen's
inequality as well as putting some
details in the appendices and of course
even more just implicit for learning but
General mathematical
details General visualization
motifs and then it's kind of like the
sighting of the models that will
resurface in seven
and8 with an interl
of five being the empirical evidence and
studies in the most well studied setting
just by like number of studies the human
nervous system then six with the recipe
before going too much into the details
of these specifically so it's almost
like fours a
preview plus a bunch of background like
pmdp again a
general long
existing um way to model perception and
action so it's interesting that for
really contains a lot of
general math and learning before and
after a sort of preview of where it's
actually going to be described in more
detail in seven and eight after the
recipe is shown and the evidence too
yeah definitely yeah this uh and just as
four is a kind of preview for the second
half of the textbook it's like chapters
two and three are also sort of these
four runners up to chapter four um I I
feel in that chapter two we have the low
road uh and

so we can get to active inference from
B's Rule and so but but of course before
we can get there for anyone coming to
this textbook who's interested
in um kind of cognitive computation in
Neuroscience uh it maybe maybe you have
to kind of take a the step to
say how do we how do we teach people
about this from the ground up and how do
we get their interest and so so to begin
with the basy and brain hypothesis for
example and starting with the bases role
I think you really do have to to kind of
make that point and and describe it well
but then of course we're going to have
to get to the mathematics about it and
if we present all of that
simultaneously um I don't know how well
the rest of the textbook is going to
read so we do have this kind of back and
forth between dropping this into the
mass we pulling us back out like
resurfacing like where where are we now
and then chapter 3 oh we go much further
above the surface now we're at the high
road and we're starting there with free
energy manifestation so in a certain way
I think the chapter 4 kind of brings us
like it's like oh both roads have now
met now we can now we know that we're
starting with B's role from the bottom
and we know
that from the top and so now we get to
get into Factor graphs and and uh
message passive and the actual like
derivation of free energy and and why we
can
justify the computation of this quantity

free energy using ginon INE equality and
all these other sort of
details um I think chapter 4 is great in
terms of I think it covers actually so
much it's sort of like it precursors so
many more advanced ideas but then it
also gathers together all the kind of
simpler components in the preceding
chapters and it's tough to find a
concise way of summarizing it though I
will

say let's look at any questions or see
if any are typed but let's just see
let's go from the bottom just if there's
new

ones

okay

okay oh wow how how are games related to
active in generative
models okay that's someone thinking
about how to to have a playable
generative

model also another

another well one one reading of of how
games are related would

be we could model games like Checkers
chess go tic-tac-toe has been done Game

Theory Etc then there's the point that

Jana made there about the

visualization and then another Avenue

that some people are exploring like

Pablo FM this was like experiential actm

games to where we based around ontology
terms and

and um role

playing and then also Pablo has has made
the three

3D adventure of

curiosity

game any other thoughts on how active relates to

games okay I think yeah I think they immensely relate to games I just want to say that really quickly like uh you know in the in the same way that because I don't know how generally the person means that question but I I also think of it as in the way that we can try and understand any entity in its environment whether it be you know a

neuron within a cortical layer or it be Mario jumping on uh enemies and getting mushrooms in a video game uh like what we're kind of doing is is that this all relates to the idea of studying p

uh and how things kind of act and and behave and and how they update themselves internally as well so um I

mean that would that would be my answer I think that you could potentially given the right like model

architecture you could have an agent who plays almost any game um I I I could easily see that being doable and and reinforcement learning more generally uh already I mean people are using that to to both create games as well as uh you know take existing games and so that's that's been done and so you can very much do that with with active inference as well um and then the fun part is if you do both uh other methods in reinforcement learning and then

Benchmark those and compare those two active inference models um a fun way of doing

it so any also this made me think about
um discreet and continuous time nested
models so it's like baseball the way it
used to be played you have the inning
structure and then the discrete time
with the pitches and any amount of time
could it collapse between the two
um events and then there's the timed
games like continu

time and then when more recently they
added the the shot clock or pitch clock
to baseball then that's like there's a
discrete structure of of events in at
bats and innings but then also it's like
at the lowest level there's a continuous

[Music]

duration it's not exactly the same thing
but it's just kind of like you can have
dis descri nested with IND discreet
continuous nested within continuous or
the Hybrid models that are in
chapter8 and then a common hybrid model
for the folk psychology and for the
updated baseball timing is the
continuous on the proximate level and
then the discreet at a slower
level yeah that's a nice Point like with
with baseball on the one hand there can
be kind of a realtime visual process say
you're a batter for
example um there can be a a sort of
continuous realtime processing
visualization for you of of what the
pitcher is going to do what kind of like
throw that they're going to do and uh so
you might be processing all that in real
time but you could almost consider like
the classification of oh they're going

to pitch it this particular style or
this style like and then you can make
that as kind of a more discreet
classification choice of like that's
what they going to do more exactly so
you can have definitely that sort of
interplay based on how you do the
nesting in the model between these like
continuous uh observation inputs or
variables and and making these kind of
discrete choices and the other way
around I think could work as
well what would you say does active
inference require SL describe let's
separate it into two
questions what would you say for these
two
questions does it require observable
behaviors does it describe observable
behaviors that's if we take the
theoretical presuppositions in the first
half of the textbook to be true or at
minimum entertain them then I would
say from one angle no they don't require
observable behaviors but then that I
mean that's really going to get into
like kind of philosophical semantic
discussions of like observable to whom
right um
like is the
inside of a
rock currently minimizing free energy I
can't see it uh it's hard for me to even
say it's Behavior because it doesn't
really match up with my typical notion
of a behavior which is where I might see
something moving or varying over
time um but if you take the free energy

principle to

to if that's still standing for you then

uh yes like free energy minimization is

occurring is occurring in many many

different systems around us and so that

said if you would like

to if you as a as an experiment

scientist would then like to go on to

describe something in terms of active

inference by observing its

Behavior um then it kind of just depends

upon your experimental methodology right

are you are you

studying um are you studying a mouse in

a test

experiment uh like that that that more

easily is observable in the in the sense

that I think this question is coming

from uh then

again we also observe tons of like

social human behaviors where it's

immensely more difficult to experiment

and see how active inference works there

but if you take free energy if you agree

with the idea of free energy

minimization then you can start to kind

of

infer uh use that as an example so to

speak and kind of make inferences about

how you think the whole society of

people behaves or or try and explain its

Behavior based upon that in active

inference terms and from there you could

attempt to model it um but anyway I

think I'm getting further away from the

original question yeah I mean it's it's

you're right though does it require

observed behavioral I mean if you're

observing

something every phenotype can be seen as

a behavioral you know that's the leg

length behavior of that time and scale

and system but like without the

observable it's like okay you know it's

it's I'm I'm

imagining this creature's femur but it's

like but then is that a mental

observable so that does get into like C

can you you know would describe

something that wasn't there but then

what what are you really trying

to what are you really trying to do with

that

model let's delete this

one okay both the dot and the prime are

used to indicate a

derivative equation

47 yeah great question here the ticks

are used for the generalized coordinates

of motion so that is first second third

Etc

derivative that's happening these two

stacks for two paired equations Y which is

the primary observables coming in from

the Imaging for example and then $\dot{x}$ dot

which is the spatialized flow map of the

underlying neural States so just fitting

a one layer fmri model you have the fmri

raw data coming in with the observables

on y that's like the input for the

regression and then you're fitting a

rate of change on neural

activity so that dot is like a

special just reflecting that you're

modeling a sequence of observables and

an underlying flow

landscape and then both of those are
being modeled across this applied series
of derivatives for the Taylor series
yeah I think that's well
said whenever it comes to the more
strictly mathematical I'd rather not add
too many
yeah it's hard to know yeah it's all
good okay the primes are just you know
more more primes means you're just
expanding the the Taylor series
expansion further outward right by
adding more terms but then the dot is
like that rate that you're trying to
yeah to approximate and it's super it is
very general
and then it's like but if you're then
then that's something that Carl has
joked about with like this is like s say
how many layers the brain have or how
many layers of temporal depth and he'll
say like five or six or something and
then say like just because that's how
many cortical layers there are first off
but but that's always been interesting
me because the cortical layers are not
the hierarchical basilar
layers each cortical column five or
whatever layers are one hierarchical
element so I don't understand why he
describes it in terms of then the
anatomical layers but that being said
then just also
realistically higher derivatives outside
of edge cases may be less important or
there's ways to just make them go away
which is like why PID control which only
goes to two or three generalized

coordinates is enough for a lot of

Robotics

settings okay for

4.11 some previous times I know we've

discussed the italics and the

Bold

okay um

I'm not sure who wrote

this I don't know who wrote it or who

could evaluate right now but it's all

yeah so I mean equation

4.11 that's

that's like a

decomposition into a sum over

time like what you're going

from free energy to like you have that

like f_{π} term which I

believe isn't that referring to like

here's the evaluated free energy of a

particular policy at a particular time

step

um like basically like it whenever it

comes to policy inference because the

agents are always like looking at

whenever they're doing policy inference

they're looking at like they have a

planning horizon or inference Horizon

thus the t and T and t is usually

used as like it's used similarly to

similarly to T but the reason why it's a

t and not a t is because like it's a

reminder that this is the time step in

internal to the model so You' be running

a simulation for 50 time steps but if

the agent's current belief is about now

then it's just going to be t it's not

going to be T

35 it could be T 35 but it'll still

be just house agent then π being policy
so that that was my understanding of
that bit and then if the equivalent for
 S would be like here's the
belief of you know what's going on now
was

just yeah this is just just to show
this is not

accurate but then it's like but is it a
force and then it's like is that a super
deep understanding SL

metaphor where like variational free
energy is a force acting on a policy
habit or is that just a rapit hole just
a hallucination because it's not
grounded in the ontology it's just
looking at the latch

Standalone and so it just picks a a
plausible sounding
interpretation I mean I could easily
imagine

skepticism

yeah okay another good math question
let's just see if there's any okay let's
just

why is the gradient of
policy of f being zero C having policy
which follows the soft

Max not sure we could think about it
more but but it's all good but yeah if
someone has a great concise way to think
of I been at

it yeah I mean that that that term on
the right was very like that's just
policy inference that's just I mean
they've not included uh you could
include γ and E

uh in there but that's that's just the

definition of policy inference and
where you're including expected M3 energy
minimization and then on the left hand
side the upper is just the the
gradient of f for so the idea is an
agent will infer a policy that attempts
to bring free energy to zero is is yeah
how I see that if I just look at that
one image
long yeah it's possible it's like the
stationarity for policy like if you're
pulling from the best possible policy
distribution you can't do better
probabilistically than pulling from the
best policy
distribution that doesn't mean that you
win every like coin flip or something
but but you wouldn't be able to out
plan better than doing
yeah yeah you could
yeah that's one of my favorite equations
oh
yeah oh yeah policy inference just
because it's so it's fun to think about
for me um it's like oh like here's a way
of quickly relating G is like what I
think will happen versus F which is like
what I think is going on right now
versus e which are my habits which are
kind of like what I've accumulated from
the past so it's just kind of like it's
incorporating past present and future
like really explicitly addressing like
behavior and what someone does so so you
could have like super precise like heavy
habits uh you could think of like a
cigarette smoker who deals with stress
by smoking right it's like well your G

might be great your F might be great but
if your habit is like really really bad
like really strong then that's going to
overtake the computation of your final
final posterior uh posteriors over
policies such that you you go and smoke
you know I don't it's crude way of like
getting into it like that but uh I I
just kind of I look at that equation
sometimes I'm just like I don't know why
I enjoy looking at this so much and like
thinking about it anyway um but yeah we
can move on to other
question yeah like those are the two
halves of inference there's perceptual
sense making inbound
inference and then kind of like full
stop that that's on the A and then
there's on the
B policy
inference and then there's so many
questions related to that like what
are how do you learn what actions you
can take everything about habit and
learning and updating context
switching policy selection different
time Horizons different tradeoffs
different
sensitivities a lot of shared
apparatus with perception that's why it
makes sense to model them in like a
unified way and with a single imperative
which is ultimately what happens the
agents minimizing free energy on its
joint
distribution so it is getting jointly
optimized and then yet still you can
kind of hold multiple parts of

it constants and then still like zoom in
on just how how one part gets optimized
or like how policy is updated without
considering because it's factored off
from the A and B are factored off
through each other through the S markov
blanket this one pending um the someone
to write the verbal
description or writing a prompt that can
transfer these into verbal descriptions
with the ontology

okay nice a lot of good math

questions what is meaning meaning of the
Tilda

ETA is this the

ETA yeah

I I think so is that the um

that kind of doesn't that come in as
some kind of

additional cause or Force sorry I

can't I'm not exactly sure here because

I thought that'sa was this shape but

maybe that's the

uh we'll come back to

this

oh yes okay here's the the twin

dynamical equations the fmri kind of
classic equations

here's the signal and noise of the

sensor data here's kind of the dominant

Trend underlying function and the faster

frequency stochastic variation of the

underlying neural

map which could be itself noiseless as

sometimes done like there's a Kind of

Perfect noiseless neural flow landscape

and then we're getting noisy sensors

also it can there can be a noise term in

the neural layer so it's self is
probabilistic and then it's like f and g
it's just common modeling like just
because f of x is the first one and then
it's common sometimes in notation just
to use the next
letter so there's a funny joke I will
not repeat here just about like why the
vitamins are named the way that they're
named but it's like why are there you
know 12 B's but then like why does it
Skip it's not it's not an scientifically
accurate joke but it's
fine the syntax for describing
expectation is never
described box okay someone can add more
what is meant by
Niche okay here's one answer environment
of agent may be possible to modify
construct and
interact what would you say is
Niche yeah it's uh it's like it feels so
easy and tough at the same time um I
mean it's the answer given uh seems
really fair the first place my mind goes
is uh it would
be the environment of an agent uh but
then that said like to give put a little
bit more meat on the bone of why we
would use the word each rather than
environment is perhaps to emphasize the
idea that as we could think of like you
know any
organism or animal or human or otherwise
being within a niche we can think of
like an environment that
specifically like the the organism fits
or maybe

better uh is trying to
fit right um so that that's kind of how
I think about it it's like uh I I would
be very out of my Niche if I were uh
stranded on the North Pole uh without
reasonable thermal equipment uh if I
just teleported right now to the North
Pole or something right
um but to be where I am right now where
you know I have an entire kind of
ecosystem of like affordances around me
that that work well with maintaining all
these homeostats that make up the
phenotype that is me at this moment uh
that's that's a good Niche to be at I
think
um so I I don't know would you think of
a niche that way just like it's like an
environment but with like a with extra
friendlier steps to the agent or
something like that I don't know yeah
those were all good that was interesting
that it's like the organism or or system
of interest it's like it fits there and
also Fitness is calculated there so
there's kind of layer and a half of of
fitting in and then also like e logical
psychology it's like no one was saying
that organisms weren't in niches or
environments yet ecological Psychology
was was a call
to incorporate
more holistic and and kind of agentic
ways of thinking about interactions with
ecologies like well if it can't see that
wavelength then it it doesn't make that
ecological cue is
invisible um then there's I think some

of axle constant's papers have like
separated the selective
Niche from the the cognitive and
developmental
Niche and so I I think a lot of Niche
Theory makes these sub niches but but
yeah especially when you build the
generative model then it's kind of like
saying what is outside of the
territory versus what is outside or you
know what are the con what is labeled
Niche on this map and then that's
usually quite clear and
categorical this is in response to the
previous session asking about how priors
originate the first prior in addition to
self- evidencing optimism bias noted
chapter 2 I want to notion the mention
of genes as
prior I.E genetic inheritance has been
framed in active as part of the prior
belief that constitutes itself I was
unable to find it but believe it was
suggested by Paul Badcock
yeah the hierarchically mechanistic Mind
Badcock ramstead constants
priston and then perhaps more recently
but not quite
exactly um on on uh there with the
empirical biological in the bio
formatics data sets but but that'll
be
TBD here's an example of like the
sensory motor interface of the
worm and so that kind of primary
behavioral layer of interfacing with the
physical material Niche Etc like the
light coming in and the receiving the

the actions of the organism and then
there's also a slower adaptive process
happening one that can still be analyzed
with particular partition Fe Etc
renormalization and then here it's like
well these are these are other these are
beliefs about this is like a phenotype
of a population being AIC
distribution like you'd have height
distribution so but there's a lot to do
here to connect
empirically yeah nice what
what yeah that that that'll be what what
are your last thoughts on uh
four and Andrew and then slyy if you
want sure uh I think
all in all I think it's great I think it
it serves up a lot of information that
like I I would almost dare to say that
if if you had to read a chapter of this
textbook in order to get a sense active
inference and you could only choose one
chapter 4 might be the way to go and
that said it's like you can't quite
contain the whole textbook in that one
chapter you have to read the rest too
but uh it's
good
okay any other
comments
okay next time later or earlier for
chapter
4 chapter
nine a nice complement to
four square
numbers
also skipping the generative model
details to focus on on the on different

parts of the math and
science okay yeah cly go for it sorry um
yeah sorry I I I um didn't really
contribute a lot actually I'm just on
chapter two at the moment so uh it's all
just pretty new um but I um do mean that
the next one will be on chapter nine if
I heard it correctly well at this time
like next week week at this UTC time
yeah it'll it'll be but okay it's all
good it's
just going across different chapters I
mean you could even argue for the
advantages of it it's not like it's a
story where you'll you'll get spoiled by
reading the end yeah so just the earlier
time then it then we'll be next week
earlier however many hours ago chapter 4
part two just altering
times yeah it's like we have two cor uh
two cohorts running simultaneously and
then that way everyone who cannot make
the morning session on a Monday can make
but can make the evening session on the
following Monday that way so we
basically spend two weeks on every
chapter but we switch which cohort is
featured uh between Monday between a
morning and evening if that makes sense
so it's like you know right this is Cap
cohort 7 right now next we it'll be
cohort 6 at this time the following week
it'll be cohort 7 at this time so on um
but yeah it it gives everyone an
opportunity to participate in h
whichever chapter reading they want to
get
into okay okay thank you thank you so

much yeah feel free to add any written
questions or just or you know write them
in Discord but then we'll just go
wherever we go especially this many
cohorts in
okay have a good one thank you farewell